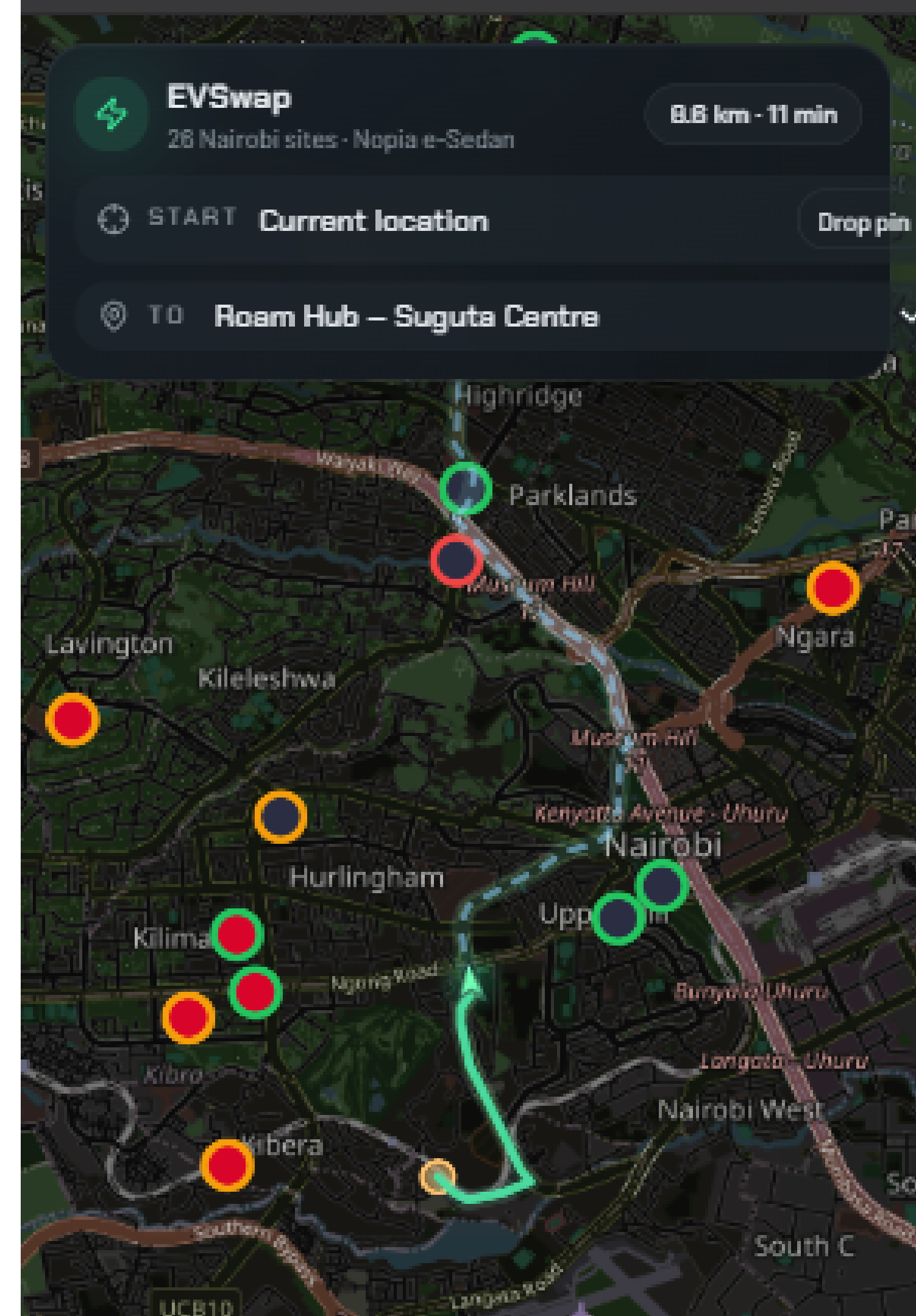


BUILT IN NAIROBI, BUILT FOR THE WORLD



A replicable AI-driven model that leverages city data to accelerate urban e-mobility.

AI X CITY HACKATHON 2026



02 | THE PROBLEM

Nairobi's transport sector is a major, growing emissions source.

- ⚡ E-mobility (buses, e-bikes, boda-bodas) is the fastest mitigation lever available today.
- 📍 Infrastructure is currently sited by **commercial logic**, not transit need.
- ⚠️ Adoption stalls wherever access lags, creating a cycle of exclusion.

TRANSPORT SECTOR
CONTRIBUTION

MAJOR

EMISSIONS SOURCE

02 | CONTEXT

Why Nairobi

A representative “high-informality, high-growth” African city. The hard case, on purpose. If it works here, it generalizes.

4.6M

residents (WorldPop 2020,
verified in this project)

136

matatu routes -
headway-based, no fixed
timetable

4,284

matatu stops
citywide

865

H3 hexagons
tessellating the city

Nairobi's matatu network runs on informal, driver-set headways rather than a published schedule. This project derives frequency directly from real GTFS trip data (Digital Matatus), not assumptions.

03 | OUR SOLUTION

An AI-driven, replicable e-mobility siting model.

We score every part of Nairobi for charging need using real population + transit data, recommending phased sites per the **Avoid-Shift-Improve** framework.



DATA-LED SITING

Leveraging real-world population and transit demand data to identify high-impact locations.



PHASED ROLLOUT

Strategic prioritization: Electrifying buses first, followed by e-bikes and boda-bodas.

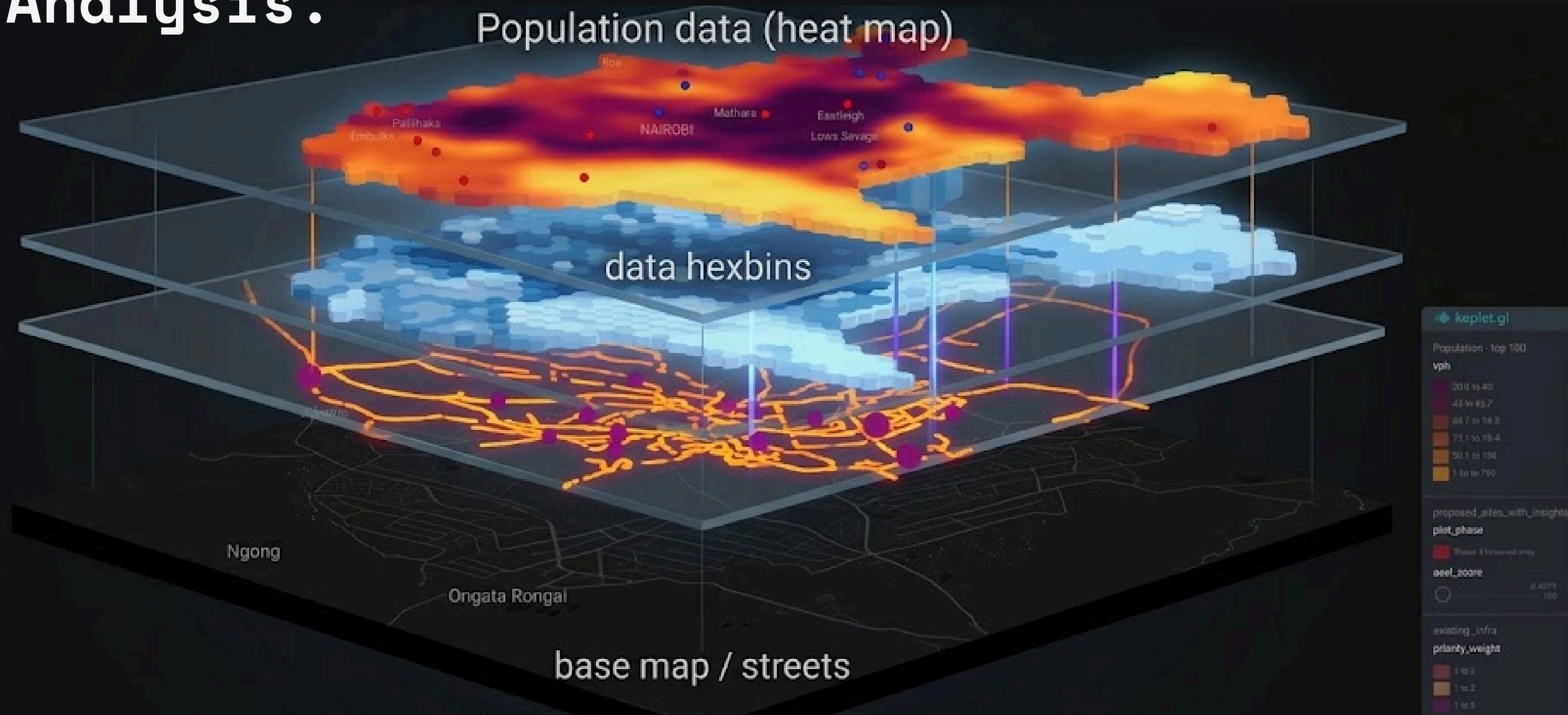


GLOBAL REPLICABILITY

Open data pipelines that can be deployed in any urbanizing city.

04 | METHODOLOGY

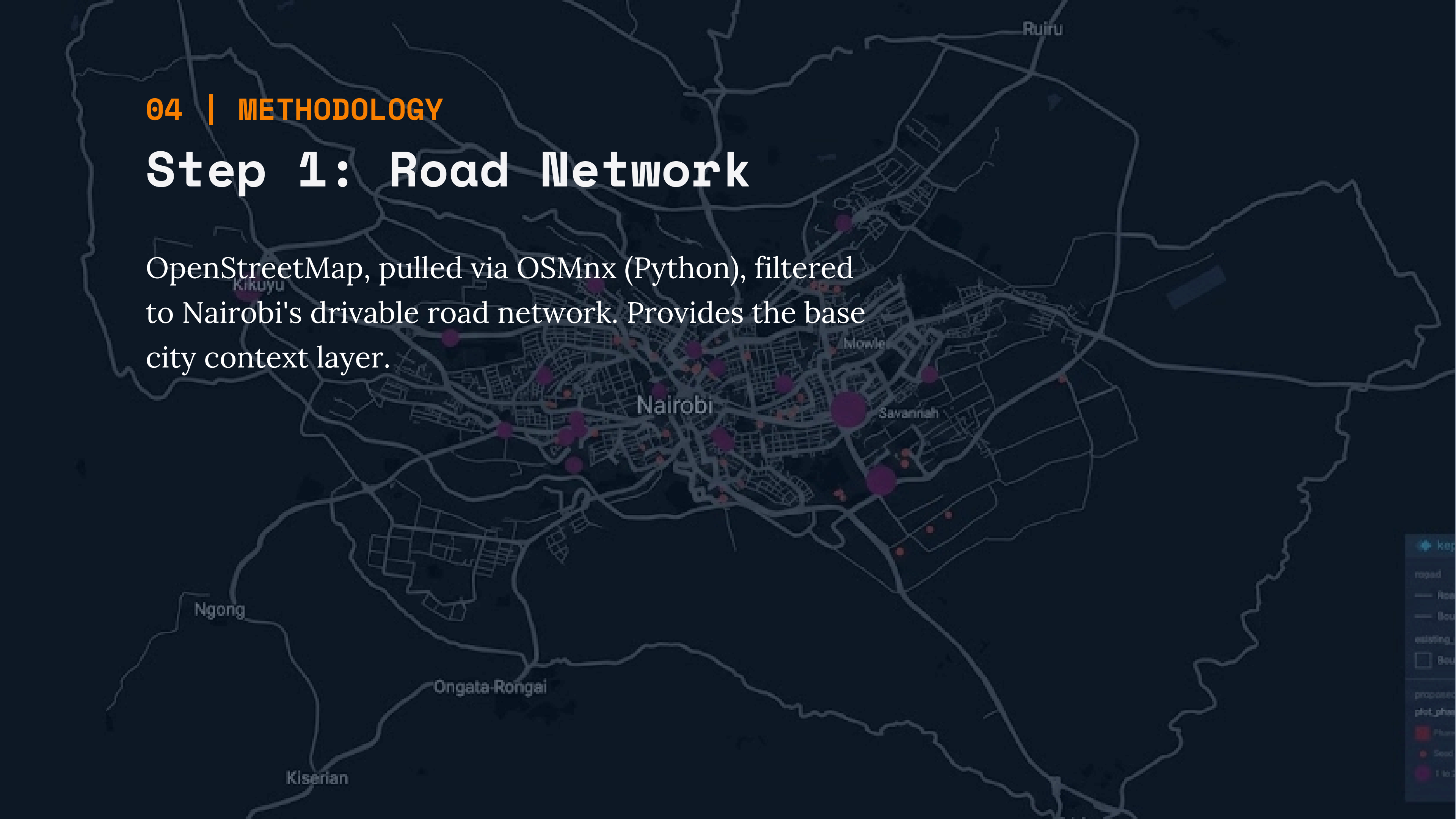
Four Step Spatial Methodology for City Analysis.



04 | METHODOLOGY

Step 1: Road Network

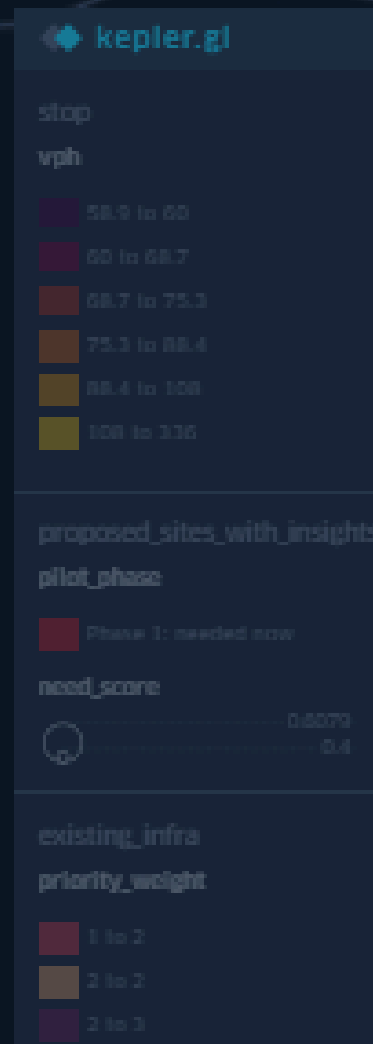
OpenStreetMap, pulled via OSMnx (Python), filtered to Nairobi's drivable road network. Provides the base city context layer.



04 | METHODOLOGY

Step 2: Population Data

WorldPop's 100m-resolution, constrained, UN-adjusted gridded population estimate for Kenya (2020). "Constrained" means it estimates population only where actual building footprints exist. Zonal-summed into each hexagon. (Total came out to 4.62M; consistent with Nairobi's known population.)



Step 3: Transport Patterns

The Digital Matatus GTFS feed for Nairobi (136 routes, 4,284 stops), sourced via TUMI Datahub. Peak/off-peak derived from the feed's own frequencies.txt defines three real time bands (AM peak 06:00–09:00, midday 09:00–15:00, PM peak 15:00–21:00), which we used directly to compute vehicles-per-hour per stop per band.



04 | METHODOLOGY

Step 4: Multi Criteria Need-analysis

A composite model per hexagon: 60% population density + 40% peak transit demand (both min-max normalized), discounted by existing e-mobility charging/swap coverage.

05 | THE AI, MADE VISIBLE

Transparent intelligence, not a black box.

01

NEED-SCORING MODEL

Multi-criteria using a transparent, auditable spatial analysis form

Each recommended site's stats are passed to an LLM for real time justification in the app.

02

GROWTH FORECASTING

WorldPop CAGR-based phased urgency to stay ahead of urban expansion.

Predictive Horizon:
2026

03

LIVE LLM INSIGHTS

Plain-language, data-grounded rationale generated in real time for every site.

"High transit demand at this node justifies a 10-bay battery swap station..."

Multi-Criteria Need-Analysis Model

```
demand_score = 0.6 × pop_norm + 0.4 ×  
transit_norm
```

```
existing_coverage =  $\sum$  priority_weight  
(point stations in hex)
```

```
need_score = demand_score × (1 -  
coverage_norm)
```

865

hexagons ranked by
e-mobility infrastructure need

**Nairobi's highest-need
e-mobility hotspots have**

ZERO

**verified charging coverage
today.**

 CURRENT INFRASTRUCTURE

Clusters heavily around commercial and CBD sites, following high-income real estate trends rather than transit demand.

 ACTUAL MOBILITY NEED

Concentrates in high-density residential corridors and peak transit hubs where charging access is non-existent.

06 | THE GAP IS GROWING

Urban growth is outpacing infrastructure.

+21,000

Predicted new residents in the #1 need hexagon by mid-2027.

COMPOUNDING RISK

Current mobility need and fastest population growth overlap perfectly.

Without data-driven intervention, the access gap compounds every month, locking in high-emission transit patterns for new residents.

🕒 CRITICAL WINDOW: NEXT 18 MONTHS

07 | PROJECTED IMPACT

Kenya's transport sector emits 11.25M tonnes CO₂e a year.

12%

of Kenya's total GHG emissions come from transport

4.6 L/100km

boda-boda fuel use vs. 33.1 L/100km for matatus

>80%

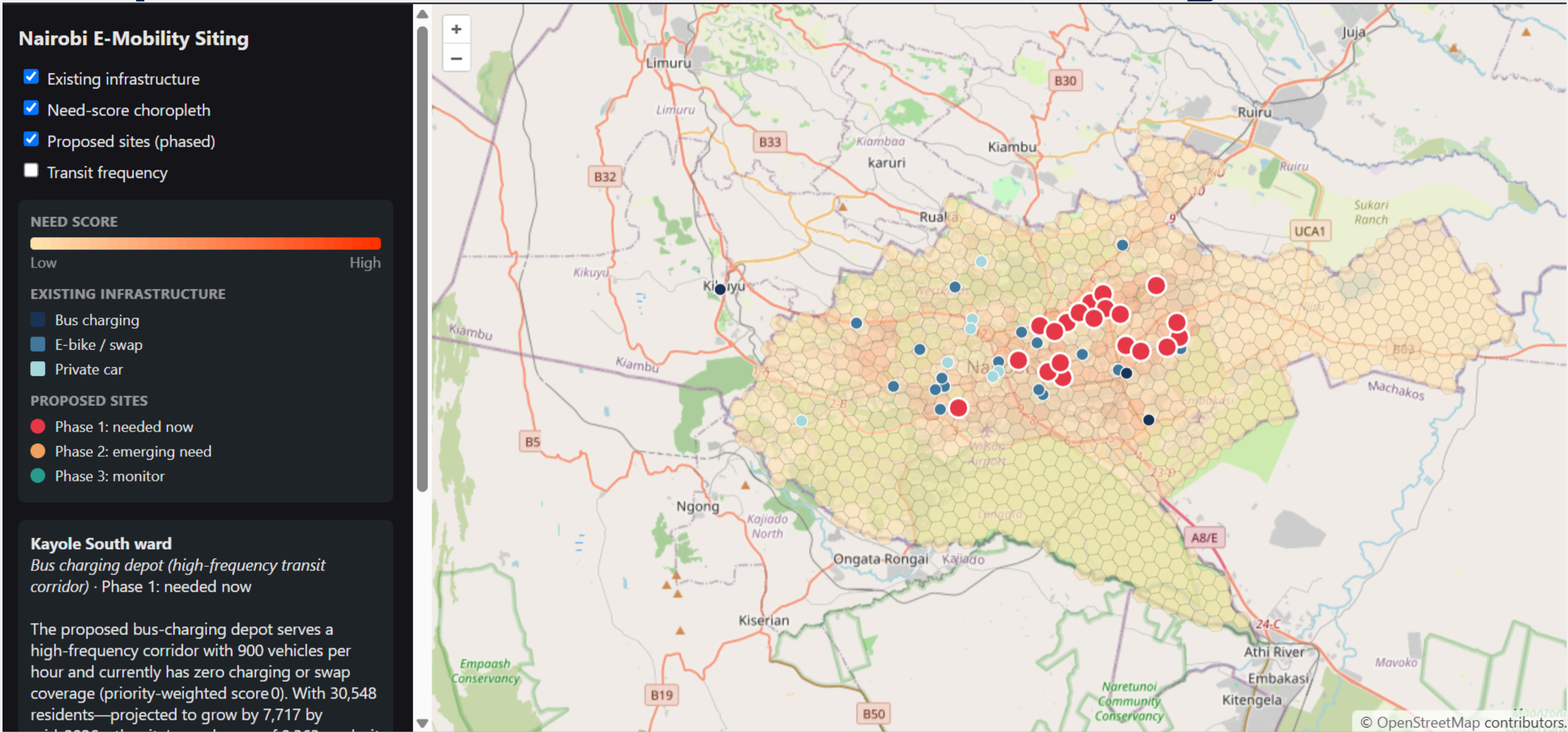
of Kenya's grid is renewable (hydro, geothermal, solar, wind)

1M+ tCO₂e

WRI's own 10-year target for Kenya's two/three-wheeler electrification

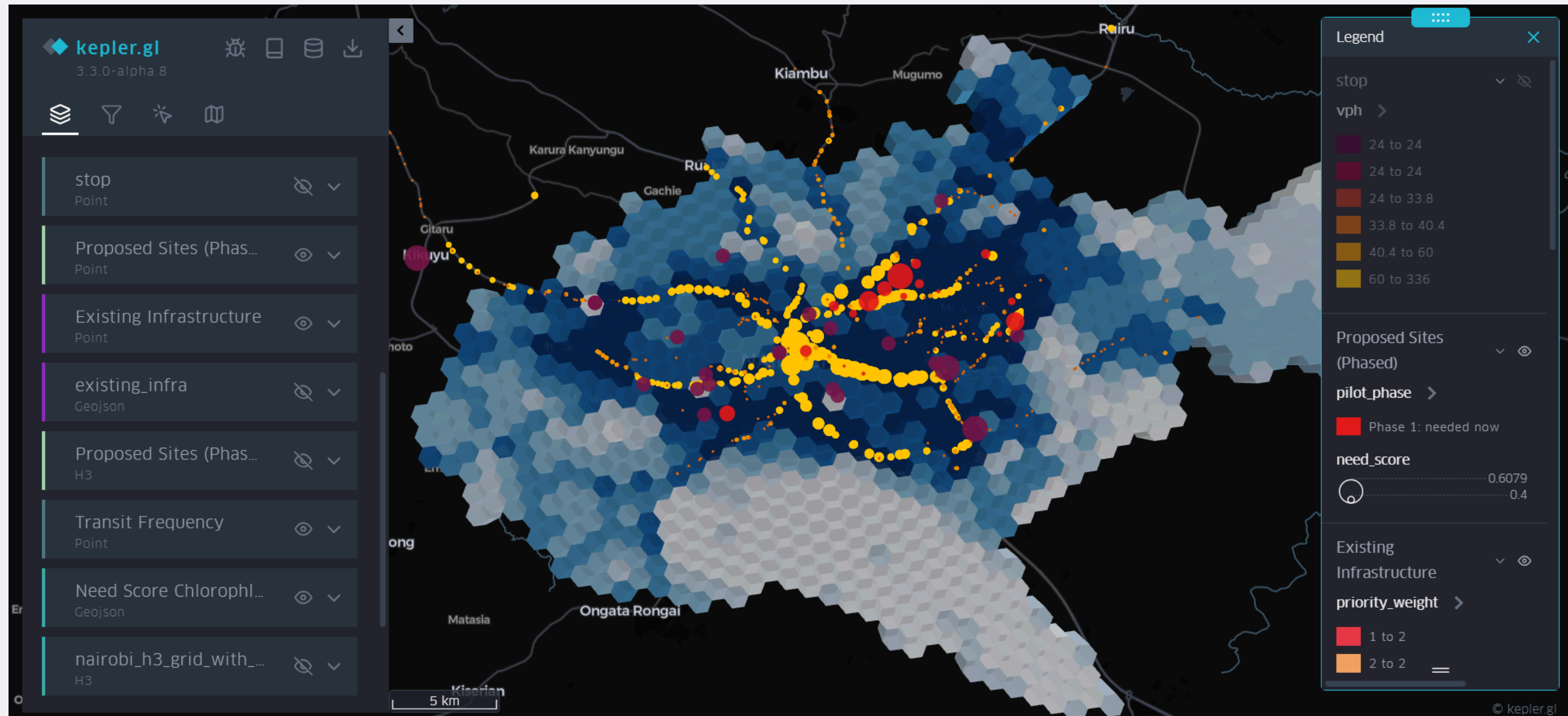
EVSwapp doesn't generate electricity or vehicles, it removes the siting bottleneck slowing this transition down. Every hexagon we get right is capacity that comes online faster."

Proposed Sites + Live AI Insights



08 | LIVE DEMO PREVIEW

Fully interactive insights Dashboard.

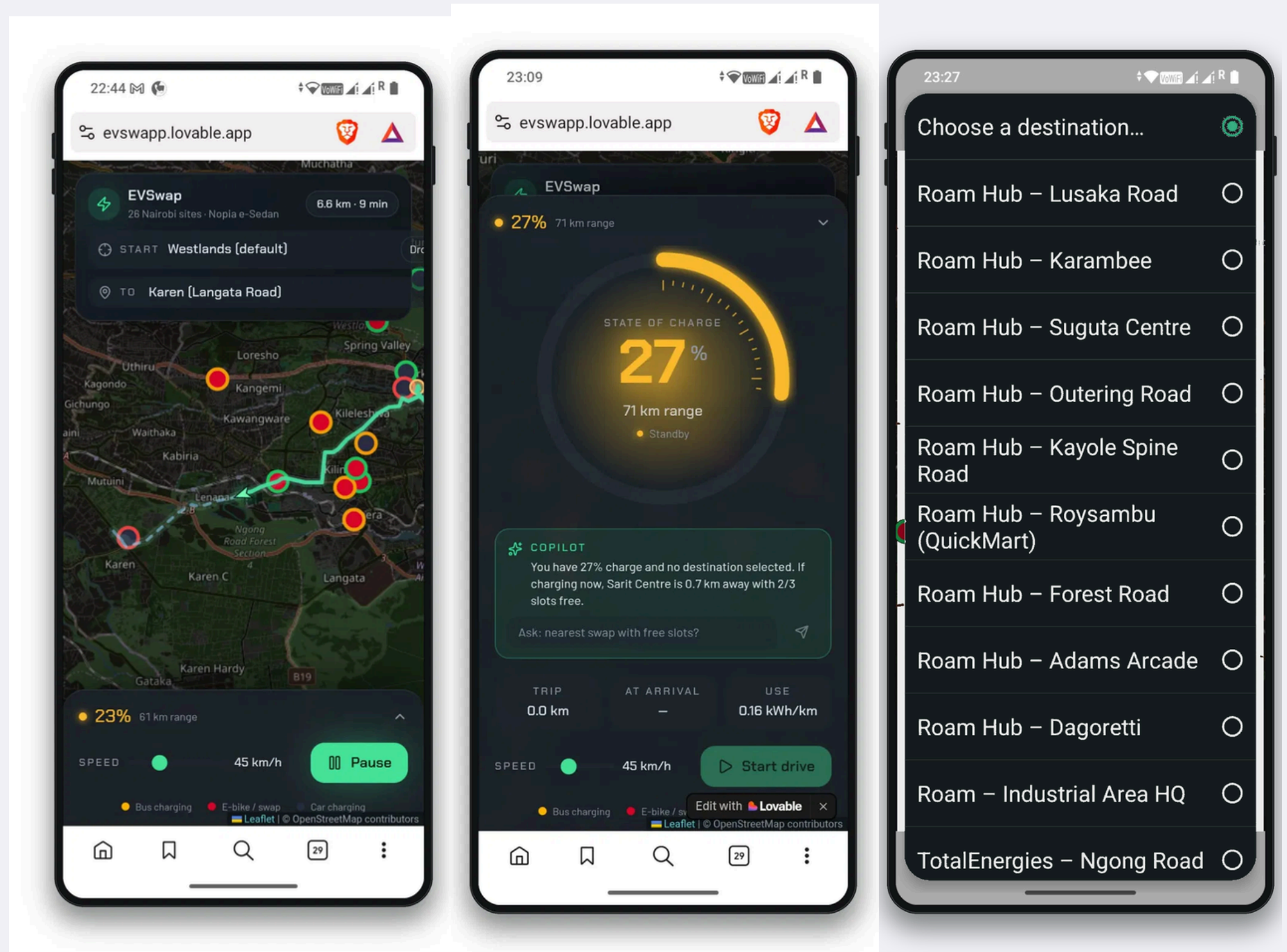


Designed for decisions, not just discovery.

LIVE PROTOTYPE AVAILABLE

09 | LIVE DEMO PREVIEW

Mobile-First EVSwapp App



Designed for decisions, not just discovery.

evswapp.lovable.app

10 | WHO THIS IS FOR

One decision layer. Multiple buyers.



CITY PLANNING & TRANSPORT

Data-driven urban policy



NATIONAL POLICY TEAMS

E-mobility roadmaps



C40 / I4C MEMBER CITIES

Global replicability



E-MOBILITY OPERATORS

Roam, BasiGo, Ampersand



REAL ESTATE OWNERS

Asset monetization



CLIMATE FINANCE

Impact investors

11 | BUSINESS MODEL & IMPACT

Here is How we Make Money

SITING-AS-A-SERVICE

Charge-point operators pay for verified, ranked, AI-explained site recommendations, turning high-risk capex into data.

MODEL: PER-REPORT / SUB

REAL ESTATE MATCHMAKING

Property owners list sites; we score them against demand and match them to operators for a conversion fee.

MODEL: BROKERAGE FEE

ADVERTISING INTELLIGENCE

High-need sites are high-traffic sites. We license footfall data and take a share of station digital-screen ad revenue.

MODEL: DATA LICENSE / REV SHARE

TARGET POPULATION

Underserved Phase-1 Hotspots

MODE SHIFT

Petrol Bodaas → E-Bikes & Buses

ALIGNMENT

C40 Transport-Emissions Goals

"Every stream monetizes the same core asset – the need-score data – sold to a different buyer for a different decision."

12 | OUR ASK & THE TEAM

We Request Support to Develop and Pilot EVSwapp With a Matched City.



Priscila Adhiambo

GIS Engineer | Urban Planner



Fauzia Omala

Software Engineer



Benard Kimani

AI/ML Engineer



Chibuike Cletus

Project Manager



Lynne Adhiambo

Urban Designer

Built in Nairobi, Built for the World.